End-to-End AI Product Management Framework - Master Summary Booklet

1. Orientation

This booklet explains a connected way to manage a product from first discovery to retirement, and shows where AI changes the questions you must ask. It is a decision framework, not a fixed process: use what the situation needs and skip the rest.

Audience. Product managers, aspiring PMs, and cross-functional leads who want one reusable mental model for AI and non-AI products.

Companion deliverables. The booklet explains, the interactive flowchart navigates, and the practical resources operationalise. All three use the same logic and terminology.

1.1 Central thesis

End-to-end product management is the coordinated practice of finding valuable problems, defining a product, designing it, gathering evidence that it works, delivering it responsibly, and improving it for as long as it lives. For AI products the responsibilities are the same, with extra questions:

- Is AI appropriate for this problem at all?
- What can the system reliably do, and when might it fail?
- How is uncertainty communicated to users?
- What safeguards and human oversight are needed?
- What evidence justifies release, and how is behaviour monitored afterwards?

The goal is not to put AI in every product. It is to make sound decisions and use AI only where it adds value.

1.2 Six concerns that must stay connected

Concern	Question it answers
Intent	What problem are we solving, for whom, and why?
Definition	What outcome and behaviour are required?

Concern	Question it answers
Design	How does it work for users, including when things go wrong?
Evidence	How will we know whether it works?
Delivery	What must be true before release?
Learning	How do we measure real-world results and improve?

Break any link and problems follow: requirements without objectives breed unnecessary features; AI without failure handling creates bad experiences; launches without evaluation expose users to avoidable harm; products without monitoring decay unnoticed.

1.3 PRD versus end-to-end management

A Product Requirements Document (PRD) defines what a product or feature should achieve, for whom, within what scope and constraints, and with what acceptance criteria. It is one important document. It does not replace discovery, evidence, design, evaluation, release, monitoring or learning. Treat a finished PRD as an input to those activities, never as proof that the product is ready.

1.4 Nine questions to ask at every decision point

1. What are we trying to achieve?
2. What do we currently know?
3. What remains uncertain?
4. What decision must be made?
5. What evidence would support it?
6. Which risks and trade-offs matter?
7. What action follows?
8. How will we know it worked?
9. What happens if the outcome is unsatisfactory?

1.5 How to use and adapt this framework

Before applying it, answer: What kind of product is this? Is AI involved, and where? What is the consequence of failure? What evidence already exists, and what is uncertain? How reversible are the big decisions? What obligations apply? What decision comes next? Then choose the minimum useful level of documentation and control. Section 11 shows how rigor scales with risk.

1.6 Labels used in this booklet

To keep knowledge honest, content is labelled where status matters: **framework knowledge** (general practice), **illustrative example** (explains a concept, not a universal rule), **recommendation** (a practice that may help, needing your own validation), and **open question** (cannot be settled from the material). The running example is called the *Illustrative AI Customer Support Project*; it is used only to explain ideas and does not imply every product needs its features.

2. The Integrated Architecture

The framework has two complementary views. The **seven study areas** describe responsibilities; the **five layers** describe the architecture and the decisions that sit between them. Neither is a rigid sequence.

A useful explanatory backbone is: **Discover → Define → Design → Evaluate → Experiment and Deliver → Launch → Monitor and Improve**. In practice, discovery can revise strategy, requirements can expose safety concerns, design can reveal missing requirements, evaluation can force a narrower scope, and production monitoring can reopen any earlier decision.

2.1 The five layers

Layer	Central question	Typical outputs	Decision gate
1. Product Intent and Strategic Direction	Right problem, right users, worthwhile reason?	Problem statement, research synthesis, opportunity assessment, vision, assumption register	Enough justification to invest in defining and investigating a solution?

Layer	Central question	Typical outputs	Decision gate
2. Product Definition and Experience	What should the product achieve and how do people use it?	PRD, backlog, acceptance criteria, journeys, flows, prototypes	Is intended behaviour clear enough to guide build and evaluation?
3. AI Capability, Behaviour and Responsibility	What should the AI do, never do, and how are risks managed?	Behaviour specs, capability boundaries, controls, escalation rules	Are capability, limits, safeguards and accountability fit for the intended use?
4. Evidence, Evaluation and Controlled Delivery	What evidence says it is ready for its intended use?	Evaluation plan, results, metric definitions, experiment plan, release checklist	Is the evidence sufficient for this release given the risks?
5. Production Learning and Lifecycle	Is it still delivering value, and what changes next?	Monitoring plan, incident records, improvement backlog, re-evaluations	Continue, improve, restrict, roll back, or retire?

A decision to proceed never means uncertainty is gone; it means the remaining uncertainty is acceptable for the next step or can be investigated properly.

2.2 How layers and study areas map

Study area	Primary layer	Also feeds
01 Discovery and Strategy	1	2, 4, 5
02 Requirements and Product Definition	2	3, 4
03 AI Design and Responsible AI	3	2, 4, 5
04 UX and Interaction Design	2	3, 4
05 AI Evaluation and Metrics	4	2, 3, 5
06 Experimentation, Delivery and Release	4	1, 5
07 Monitoring and Continuous Improvement	5	1, 2, 3, 4

This mapping is a recommended reading of the source material, not a fixed taxonomy.

2.3 Feedback loops to keep visible

- **Evaluation to design:** failed tests lead to redesign, a narrower scope, added controls, or a decision not to release.
- **Usability to requirements:** testing exposes missing failure paths and unclear requirements.
- **Experiments to priorities:** results change what gets built next.
- **Production to discovery:** real usage reveals new needs and failure modes.
- **Change to re-evaluation:** a changed model, prompt, data source or workflow may require fresh evaluation and release approval.

2.4 Decision gates without rigid stage gates

Use a gate only when it prevents premature commitment, clarifies accountability, or manages material risk. A good gate states the decision, the evidence required, the acceptable uncertainty, who is accountable, and the possible outcomes: proceed, proceed with conditions, investigate further, revise, pause, roll back, or stop.

3. Study 01: Product Discovery and Strategy

Why it matters. Discovery decides whether a meaningful problem exists and whether pursuing it is justified. It stops teams building a technically interesting solution for a need that is not valuable enough.

Decisions it supports. Proceed, investigate further, change direction, or stop; and what the MVP should be.

3.1 Core questions

Who has the problem? What are they trying to do? What gets in the way? How do they cope today? What evidence shows it matters? What alternatives exist? Is it strategically and commercially sensible? Which assumptions are untested?

3.2 Key concepts

- **Problem framing:** describe the underlying user or business problem without assuming a particular feature is the answer.
- **User and stakeholder research:** goals, constraints and existing workflows of affected people; personas and jobs-to-be-done where they help.
- **Evidence quality:** how information was obtained, how representative it is, and what conclusions it can legitimately support.
- **Vision and value proposition:** the future state and why the product is useful, for whom.
- **Opportunity prioritisation:** compare options on user value, strategic fit, feasibility, viability, desirability, cost, uncertainty and risk.
- **MVP:** the smallest scope that tests key assumptions or delivers meaningful value. It is not simply "as few features as possible"; state what it is meant to validate and why items are in or out.

Business viability and operating economics (where relevant). Connect product value to viability: who benefits and who pays, what it costs to build and to keep running (including usage-based AI costs, support and maintenance), and whether the value justifies that ongoing cost. Treat cost and benefit figures as estimates until measured, and revisit them as evidence arrives.

Data readiness and governance (where relevant). Check early whether the data the product depends on exists and what is known about its quality, access rights, provenance, privacy, retention and governance. Unconfirmed data readiness is an

assumption to log, not a given.

3.3 Evidence discipline

Personas, market claims and customer insights are not validated facts unless their evidence supports that status. An AI-generated persona can help explore a scenario; it does not prove such a person exists. Simulated interviews can rehearse questions; they cannot replace real interviews when the goal is to establish genuine needs.

Evidence type	What it can support
Real participant research	Claims about those participants, within sample limits
External research	Context, with source quality checked
Internal stakeholder input	Business constraints and views, not user evidence
AI-generated or simulated material	Hypotheses and rehearsal only
Assumptions	Items to validate, never findings

3.4 Outputs (only those you need)

Problem statement, research synthesis, provisional personas, current-state journey, opportunity assessment, vision, value proposition, objectives and success measures, assumption and risk register, prioritised opportunities, MVP rationale.

3.5 Where AI enters early

Ask whether AI is appropriate at all, who is affected by errors, and what must be learned before committing. These are Responsible AI questions asked during discovery, not later.

3.6 What can go wrong

- Starting with a favoured solution instead of the problem.
- Treating assumptions as facts.
- Generalising from a small or biased sample.
- Confusing stakeholder preference with user evidence.
- Defining an MVP without saying what it validates.
- Not linking findings to later decisions.

3.7 Connections

Discovery feeds objectives, scope, requirements, design priorities, evaluation criteria and release decisions. Testing and production evidence can reopen it, so treat discovery as an ongoing source of learning, not just a first phase.

4. Study 02: Requirements and Product Definition

Why it matters. Definition turns an understood problem into a clear statement of what the product must achieve and how its behaviour will be judged. It links strategy to design, build, evaluation and release.

Decisions it supports. Scope and exclusions, priorities, whether requirements are testable enough to build, and when requirements must change.

4.1 Building blocks

- **Goals and measurable outcomes**, with scope and deliberate exclusions.
- **Functional requirements** (what it does) and **non-functional requirements** (performance, reliability, accessibility, security, privacy).
- **User stories and use cases** to discuss and prioritise value.
- **Acceptance criteria:** conditions under which a requirement counts as satisfied.
- **Business rules, constraints, dependencies, assumptions, risks, open questions.**
- **Backlog management:** keep work organised, prioritised and refined as learning arrives.
- **Traceability:** useful links from objectives to requirements, tests, evidence and decisions.

4.2 PRD in its place

A PRD typically records goals, scope, users, requirements, constraints and acceptance criteria. Research, experimentation, detailed AI evaluation, release management, monitoring and continuous improvement need their own processes and artifacts. The PRD should connect to them rather than stand in for them.

4.3 Requirements for AI capabilities

An AI requirement often needs more than a desired output. Depending on the capability, state:

- expected behaviour in ordinary conditions;

- supported and unsupported inputs;
- context and state needs;
- handling of ambiguity and uncertainty;
- unacceptable outcomes;
- escalation and fallback behaviour;
- privacy and safety constraints;
- evaluation scenarios and thresholds;
- monitoring needs and conditions that require re-evaluation.

Describe observable behaviour wherever possible, so someone else can test it.

Technical and operational dependencies (where relevant). Record what the product needs from engineering, platform, security, integration, infrastructure, data and operations teams, with an owner and a clear handoff for each. Unowned dependencies are a common reason that agreed requirements cannot be delivered or sustained.

4.4 Illustrative example (not a universal rule)

In the Illustrative AI Customer Support Project, a requirement might concern detecting a repetitive, unproductive conversation and offering a human handoff. The reusable lesson is not "every product needs loop detection" but "define the trigger, the expected behaviour, the handoff, and how you will test it."

4.5 What can go wrong

- Requirements too vague to test.
- Features disconnected from outcomes.
- Acceptance criteria covering only ideal scenarios.
- Missing non-functional requirements.
- Unrecorded assumptions and dependencies; duplicate or conflicting requirements.
- Requirements with no evaluation approach.
- Treating a completed PRD as proof of readiness.

4.6 Connections

Requirements guide design, AI behaviour, engineering, testing, release planning and monitoring. Evaluation may show they need clarifying. A requirement is strongest when its purpose, expected behaviour, acceptance conditions and evidence are connected.

5. Study 03: AI Product Design and Responsible AI

Why it matters. The question is not whether AI *can* do a task, but whether the capability is useful, reliable enough for its context, and appropriately controlled. Responsible AI is woven through every stage, not completed once as a checklist.

Decisions it supports. Whether to use AI at all, what it may and may not do, what safeguards and oversight are needed, and whether risks are acceptable for release.

5.1 Is AI the right tool?

Compare AI with deterministic rules and with human judgement. Deterministic logic may be better where behaviour must be exact and explainable; a human may be better where consequences are high or context is subtle. Reconsider AI if evidence shows it is unreliable, needlessly complex or unsuitable.

5.2 Capability boundaries

Define what the system is expected to handle and what is outside intended use. A conversational system, for example, may answer supported questions but must not imply it can reliably resolve every request. Boundaries shape user messaging, requirements, tests and escalation. Keep **intended behaviour** separate from **demonstrated capability**.

5.3 Context and memory (only where relevant)

Persistent memory is not universally needed. Where context retention helps, define what is retained and why, for how long, when it is updated or discarded, how stale or contradictory information is handled, which privacy limits apply, and how retention quality is evaluated.

The same discipline applies to any data an AI capability uses: where it comes from (provenance), its quality, who may access it, privacy and retention limits, and who is accountable for governing it. These are requirements to define and test, not assumptions to leave implicit.

5.4 Uncertainty and failure handling

For each relevant capability work through seven steps: how the problem arises; how it is detected; how the system responds; what the user is told; whether the interaction can recover; whether a human is needed; and how the incident feeds future evaluation.

5.5 Responsible AI as integrated practice

Relevant considerations include privacy, security, fairness, transparency, reliability, human accountability, access control, safety and foreseeable misuse, and auditability. Which controls apply depends on the product, users, obligations and cost of failure.

Stage	Responsible AI actions
Discovery	Identify affected users and consequences; ask whether AI is appropriate
Requirements	Define acceptable and unacceptable behaviour; set privacy, security and oversight needs
Design	Communicate boundaries; design uncertainty and failure paths; avoid interfaces that invite unjustified trust
Evaluation	Test realistic and adverse conditions; assess important failure categories; record limitations
Release	Confirm controls are in place; weigh unresolved risks; set ownership and incident response
Production	Monitor failures and impact; review incidents; reassess safeguards after changes

5.6 Making safeguards testable

A safeguard is more than "the AI must be safe." Specify the risk addressed, the trigger, the required system behaviour, the user experience, the test method, the acceptable result, the fallback if the safeguard fails, the owner, and a production signal that may indicate failure.

5.7 Human oversight that actually works

Give human involvement a defined purpose: the trigger for intervention; the responsible person or team; the information they receive; the actions they may take; how the handoff is communicated; what happens if nobody is available; and how effectiveness is assessed. A vague promise of "human in the loop" is not a safeguard.

5.8 What can go wrong

- Assuming AI behaves consistently across all inputs.
- Treating safeguards as aspirations.
- No defined behaviour when confidence is low.
- Oversight with no workable handoff.
- Ignoring privacy of contextual data.
- Testing normal cases while missing consequential edge cases.
- Assuming an explanation of an output proves it is correct.

5.9 Connections

AI design affects requirements, UX, evaluation, release and monitoring. Evidence from testing or production can change the capability, the safeguards, or the decision to use AI.

6. Study 04: User Experience and Interaction Design

Why it matters. UX decides how users meet, understand, control and recover from the product. Quality includes how well people handle unsuccessful interactions, not only how smooth the successful ones are.

Decisions it supports. Which journeys and failure paths to design, how limits and uncertainty are communicated, how people reach a human, and what usability evidence is enough.

6.1 Key concepts

- **User journeys and interaction flows:** the steps, decisions, responses and alternative paths of a task.
- **Information architecture, wireframes and prototypes:** explore structure and behaviour before full build.
- **Conversation design** (for conversational products): how the system interprets requests, clarifies, responds and manages context.
- **Error recovery, accessibility and usability.**
- **User control and feedback:** ways to correct the system, report problems or express dissatisfaction.
- **Communicating AI limits and uncertainty** without hiding it or overstating confidence.

6.2 Happy paths are not enough

Design must cover the paths below, and each should connect to a requirement and an evaluation scenario rather than being left to implementation chance.

Situation	Design question
Ambiguous input	How does the product clarify instead of guessing?
Misunderstood intent	How can the user correct it quickly?
Missing information	What is asked for, and how?
Unavailable capability	How is the limit stated honestly?
Uncertain or wrong AI response	How is uncertainty shown, and how can the user challenge it?
Upload or dependency failure	What recovery path exists?
User asks for a human	How easy is it, and what context carries over?
Handoff itself fails	What is the fallback?

6.3 Human handoff experience

Where human help is part of the design, users should find it without a fight, understand what happens next, and not have to repeat themselves. Define what information transfers and what is communicated if no one is available.

6.4 What can go wrong

- Designing only the ideal journey.
- Unclear error messages.
- Hiding AI limitations or uncertainty.
- Making human help hard to reach.
- Prototyping a flow without validating usability.
- Treating a polished interface as proof of effectiveness.

6.5 Connections

UX puts requirements and AI behaviour in front of users. Usability testing can reveal missing requirements, unsuitable AI behaviour, or even a need to change the product concept.

7. Study 05: AI Evaluation and Product Metrics

Why it matters. Evaluation produces the evidence behind decisions to improve, release and keep using a product. For AI it must cover both product outcomes and the behaviour of specific AI capabilities.

Decisions it supports. Whether performance is adequate, what to fix, whether to release, and when to re-test.

7.1 Three measurement questions

Question	Asks
Product outcomes	Does the product create the intended value?
AI behaviour	Does the capability behave as expected across relevant inputs and conditions?
Operational performance	Does it stay reliable and usable in its real environment?

One does not answer the others. Accurate outputs can still fail users, and a good aggregate number can hide a serious failure affecting a smaller group.

7.2 Metric categories to keep distinct

- **Outcome metrics:** is the user or business outcome achieved?
- **Behavioural quality metrics:** how well does a capability behave?
- **Operational metrics:** availability, latency, reliability, cost.
- **Guardrail metrics:** conditions that must stay within limits.
- **Learning metrics:** did an experiment answer its question?

7.3 Designing AI evaluation

Include representative normal cases, difficult and ambiguous cases, edge cases, unsupported requests, failure scenarios, varied user and input conditions, and potentially harmful errors. Distinguish false positives from false negatives where they apply. Document test-data limitations and how far results generalise to production. Use reference labels or answers where appropriate, human or expert review where automated measures fall short, and regression tests whenever something changes. Test coverage and method depend on the capability and its risk; not every evaluation needs a numerical threshold, but qualitative criteria must be explicit.

7.4 Operational definition of a metric

For each important metric, record as appropriate: meaning, numerator and denominator, calculation method, data source, measurement window, baseline, target and its rationale, known limitations, owner, and the decision it informs. Similar names may not mean the same thing: resolution time and handling time can measure different things, so a drop in one does not prove an equal drop in the other.

7.5 Targets are not results

Status	Meaning
Proposed target	A desired level, not evidence
Hypothesis	A predicted improvement, not an outcome
Test result	Measured under test conditions
Production result	Observed in real use
Independently validated	Confirmed by a separate party or method

A 90% accuracy target does not show the system achieves 90%.

7.6 Evaluation must inform a decision

Link each important test to a requirement, scenario, expected result, measurement, threshold or judgement rule, failure response, decision owner and re-evaluation trigger. A test with no expected result is hard to read; a result with no decision rule may not lead to action.

7.7 What can go wrong

- Choosing metrics because they are easy to measure.
- Reporting aggregates without inspecting key failure categories.
- Unrepresentative test examples.
- Changing metric definitions without recording it.
- Thresholds with no rationale.
- Treating benchmark results as a production guarantee.
- Measuring activity instead of user or business outcomes. *(The source text for Document 1 breaks off at this point; this last item is completed from the master instruction's intent and should be confirmed.)*

7.8 Connections

Evaluation informs requirements, AI safeguards, experiments, release readiness and monitoring. Failed evaluation may require redesign, a narrower scope, extra controls, more testing, or a decision not to release.

8. Study 06: Experimentation, Delivery and Release

Source note: Document 1 was truncated before this study. This section is built from the master instruction's topic list and Document 2's experiment and release logic. Treat it as a synthesis to review, not a verified summary of the original study.

Why it matters. Experiments and staged delivery turn hypotheses into evidence, and release decisions turn evidence into action under controlled risk.

Decisions it supports. Whether to run an experiment, what the MVP should validate, whether the evidence supports release, how to stage it, and when to roll back.

8.1 What experiments can and cannot show

An experiment should answer a meaningful question. It can compare an intervention with a baseline under stated conditions. It cannot establish things outside its population, duration, traffic or measures, and it should never be used to justify avoidable exposure to harmful behaviour. Do not prescribe a method (such as A/B testing) that does not fit the product's scale, risk, traffic or constraints; alternatives include staged rollouts, expert review, shadow testing, and structured pilots.

8.2 A practical experiment plan

1. Hypothesis and rationale
2. Population or context
3. Intervention and comparison
4. Primary outcome
5. Guardrail measures
6. Method and duration
7. Data and evaluation approach
8. Limitations
9. Decision criteria
10. Follow-up action

8.3 MVP validation and staged delivery

Use the MVP to test the assumptions named in discovery. Stage delivery so each step limits exposure and generates evidence: for example internal use, a small cohort, then broader rollout, with defined criteria for moving between stages. Backlog refinement, engineering collaboration and stakeholder communication run alongside.

8.4 Release readiness is evidence-based

Dimension	Readiness question
Product scope	Is the intended scope clear?
Requirements	Are essential acceptance criteria satisfied?
AI behaviour	Has it been evaluated under relevant conditions?
Safeguards	Have material risks and controls been assessed?
User experience	Can users understand and recover from important failures?
Operations	Are monitoring and incident response ready?
Human oversight	Can required handoffs really be completed?
Evidence	Are results reliable enough for this release?
Accountability	Are ownership and escalation clear?

Dimension	Readiness question
Reversibility	Can it be restricted, disabled or rolled back?

A checklist supports a decision; it does not make it. The standard depends on product and risk. Possible outcomes: proceed, proceed with conditions, investigate further, revise, pause, roll back, or stop.

Where relevant, readiness also covers: technical, security, integration and operational dependencies resolved and owned; users able to adopt the product (training, workflow changes and support in place); a running cost that is understood and sustainable; and named post-launch owners with the authority to restrict, roll back or retire.

8.5 Rollback, fallback and mitigation

Plan these before launch: how to switch off or restrict a capability, what the fallback experience is, who can trigger it, and how users are informed. Go-to-market planning applies where relevant, and post-release validation checks that expected outcomes actually appear.

Adoption and change management (where relevant). Delivery is not the same as adoption. Plan for the people affected: training, changes to existing workflows, communication, and ongoing support. Check whether the engineering, platform and operations handoffs needed to run the product after launch are agreed.

8.6 What can go wrong

- Experiments with no decision attached.
- Methods unsuited to traffic or risk.
- Launching on a checklist rather than evidence.
- No rollback path.
- Treating launch as the end of the work.

8.7 Connections

Experiment results can change priorities and reopen discovery. Release evidence comes from evaluation; release decisions hand over to monitoring.

9. Study 07: Monitoring and Continuous Improvement

Source note: Document 1 was truncated before this study. This section is synthesised from the master instruction and Document 2 (Layer 5). Review before relying on it.

Why it matters. Launch is the start of production learning. Without monitoring, an AI product can drift or degrade unnoticed, and real user behaviour never reaches the team.

Decisions it supports. Continue unchanged, improve, restrict, roll back, or retire; and what to prioritise next.

9.1 What to monitor

Area	Examples (choose what fits the product)
Product outcomes	The user and business results defined in discovery
Quality and reliability	Error rates, failure categories, regression signals
AI behaviour	Output quality, unsupported-request handling, drift in inputs or behaviour
Escalation and fallback	Handoff triggers, success of handoffs, fallback usage
User voice	Feedback, complaints, correction behaviour
Operations	Availability, latency, cost

9.2 Incident handling and root cause

Define how failures are detected, who responds, what triggers restriction or rollback, and how incidents are reviewed. Each AI failure should feed back into evaluation scenarios and, where needed, requirements or design.

9.3 Change control and revalidation

Changes to a model, prompt, data source or workflow can alter behaviour. Decide in advance which changes require renewed evaluation and release approval, scaled to risk.

9.4 Continuous prioritisation

Fold monitoring evidence, feedback and incidents into one improvement backlog, ranked by user impact, risk, effort and learning value. Retirement or replacement is a legitimate lifecycle outcome when the product no longer delivers acceptable value.

Post-launch accountability. Name, before launch, who owns monitoring, who receives escalations, who may order corrective action, a restriction or a rollback, and who decides on retirement. Include whether the product still earns its ongoing operating cost and whether users are actually adopting it. A signal with no owner with authority to act does not protect anyone.

9.5 Closing the loop

Production finding	Area to reconsider
Users need something not served	Discovery
Behaviour contradicts a requirement or the requirement was wrong	Requirements
Safeguard failed or AI unsuitable	AI design and Responsible AI
Users cannot recover from a failure	UX
Metric misleading or test set unrepresentative	Evaluation
Rollout process inadequate	Experimentation and release

9.6 What can go wrong

- Monitoring with no owner or action rule.
- Dashboards that track activity, not outcomes.
- Incidents closed without updating tests.
- Changing the model or prompt without revalidation.
- Never retiring a product that no longer earns its place.

9.7 Connections

Production evidence can reopen any earlier decision; this is the main feedback loop of the whole framework.

10. Cross-Cutting Practices and the Decision System

Six foundations run through every layer. They are not a single chapter; each shows up wherever decisions are made.

10.1 The six foundations

Foundation	What it means in practice
Evidence and uncertainty	Record what is known, how, what is uncertain, and what would resolve it. Separate verified facts, observed evidence, research findings, estimates, assumptions, hypotheses, simulated material, recommendations and unknowns.
Traceability	Keep useful links from problem to outcome, requirement, behaviour, safeguard, test, evidence, decision and monitoring.
Proportionate risk management	Match testing, oversight, documentation and approval to consequences, users, environment and uncertainty. Avoid both under-controlling and over-burdening.
Human accountability	Say who owns decisions, reviews outputs, responds to failures and approves changes; give oversight a purpose and a workable process.
Failure and uncertainty management	For key capabilities: expected behaviour, limits, ambiguity, failure modes, detection, user communication, recovery, escalation, evaluation, monitoring signals.
Continuous learning	Treat launch as a move into production learning, and feed new evidence into priorities, requirements and evaluation.

10.2 The decision system

Every major activity should connect to a decision. Use: **Question → Activity → Evidence → Decision → Action → Follow-up**.

Element	Illustrative example
Question	Can the AI reliably spot a repetitive interaction pattern?
Activity	Define test cases and evaluate detection
Evidence	Correct detections, missed cases, false alarms
Decision	Is performance acceptable for the intended use?
Action	Improve, add safeguards, narrow scope, or proceed
Follow-up	Monitor real-world performance; review failures

This example asserts no performance level for any real system.

10.3 The traceability chain

Problem → Outcome → Requirement → Expected behaviour → Risk/Safeguard → Test → Evidence → Release or product decision → Monitoring. Adapt the detail to the work; not every feature needs every link.

Illustrative example (context retention, generic): users repeat themselves, so the outcome is less repetition without inappropriate retention. The requirement defines what is retained under which conditions; the risk is stale or inappropriate retention; tests cover missing, conflicting, outdated and should-not-retain cases; evidence is context-use accuracy and failure cases; the decision judges whether behaviour and safeguards are acceptable; monitoring reviews errors and feedback after release. This shows the method, not that any product has implemented or validated it.

10.4 A reusable decision record

For consequential or hard-to-reverse choices record: the decision, context, realistic options, evidence, assumptions, risks, trade-offs, decision owner, rationale, and the review trigger that would justify revisiting it. Minor choices do not need a formal record.

10.5 Risk proportionality

Context	Typical emphasis
Small, low-risk	Concise problem statement, clear scope, light requirements, basic user testing, simple monitoring
Medium complexity	Plus structured research, detailed acceptance criteria, defined evaluation scenarios, explicit dependencies, release criteria, improvement process
High-impact AI	Plus stronger evidence of need and suitability, extensive evaluation, risk and safeguard mapping, human oversight, clear accountability, stricter release criteria, incident response, revalidation after material change

These are illustrative patterns. Legal, regulatory, security and organisational obligations must also be checked.

11. Artifacts, Decision Aids and Adapting Rigor

Artifacts exist to support decisions. None is mandatory; choose by scale, risk, uncertainty and environment. Link rather than copy: an acceptance criterion should live in one place and be referenced by its test and evidence.

11.1 Artifact families

Artifact	Purpose	Supports decision	Often omitted when
A. Discovery and Strategy	Problem, users, evidence, value, assumptions	Proceed, investigate, change direction, stop	The problem is already well evidenced and small
B. PRD and Agile Backlog	Scope, requirements, acceptance criteria, priorities	Build and requirement-level validation	A tiny change needs only a ticket

Artifact	Purpose	Supports decision	Often omitted when
C. AI Features and Responsible AI	Capability boundaries, risks, safeguards, oversight	AI design, risk, release	No AI is involved
D. UX and Interaction Design	Journeys, flows, prototypes, failure paths	Design and usability validation	Interface is trivial or unchanged
E. Evaluation and Metrics	Evaluation plan, scenarios, metric dictionary, results	Improvement, experiment and readiness	Never fully; scale it down instead
F. Experimentation and Release	Hypotheses, readiness criteria, go/no-go, rollback	Controlled release	Low-risk, easily reversible change
G. Monitoring and Improvement	Signals, incident process, improvement backlog	Production management	Short-lived throwaway work

For each artifact keep in mind: purpose, the decision it supports, essential contents, inputs and outputs, links to other artifacts, when it is useful or unnecessary, and who maintains it.

11.2 Supporting decision aids

The companion resource pack (provided with this booklet) offers short, simplifiable templates:

- Assumption and risk register
- Decision record
- Requirement-to-test traceability matrix
- AI behaviour and safeguard matrix
- Evaluation scenario template
- Metrics dictionary
- Experiment plan
- Release-readiness checklist
- Monitoring and incident-response checklist

- Continuous-improvement prioritisation aid

Each states when it helps and when to simplify or skip it.

11.3 Adaptation questions

What kind of product is it? Where is AI involved? What is the consequence of failure? What evidence exists and what is uncertain? How reversible are the main decisions? What resources and obligations apply? Which decision is next? What is the minimum useful documentation and control?

Also ask, where relevant: Is the product viable to build and to keep running? Is the data ready and governed? Which technical and operational dependencies need owners? How will users adopt it? Who is accountable after launch?

11.4 Applying it to non-AI products

The framework works without AI. Skip Layer 3 controls that do not apply, keep discovery, definition, UX, evaluation, release and monitoring, and let risk set the rigor. AI-specific practices switch on only when relevant.

12. Failure Modes and Knowledge Integrity

12.1 Framework-wide failure modes and fixes

Failure mode	Treatment
Discovery disconnected from requirements	Link problem evidence and objectives to each important requirement
Untestable requirements	State observable behaviour and an assessment method
AI described without boundaries	Define supported behaviour, limits and exclusions
Responsible AI as a standalone chapter	Tie risks to requirements, UX, evaluation, release and monitoring
Vague human oversight	Define triggers, owners, handoff and fallback

Failure mode	Treatment
UX covering only success	Include error, uncertainty, recovery and escalation paths
Named but undefined metrics	Specify meaning, calculation, data and limits
Evaluation with no decision consequence	Attach a decision rule to each key test
Unclear release criteria	Set proportionate readiness conditions and accountability
Monitoring disconnected from improvement	Route findings into prioritisation and re-evaluation
Duplicated documentation	Give each artifact one purpose and link instead of copying
Strictly linear lifecycle	Show loops and allow earlier decisions to reopen
AI imposed on every product	Make AI controls conditional on relevance
Framework too heavy	Scale process to context and risk
Examples treated as universal rules	Label examples and say when the lesson generalises
Targets confused with results	Keep targets, hypotheses, test findings and production outcomes separate
Prototype confused with implementation	State the maturity of each capability
Booklet and flowchart disagree	Reconcile terms, paths and decisions

12.2 Knowledge-integrity rules

- Do not invent evidence; assumptions, targets, examples and simulated findings are never verified results.
- Do not invent completeness; if a definition, owner or threshold is missing, name the gap.
- Do not silently reconcile contradictions; say whether a resolution is confirmed or only proposed.

- A concept or prototype does not prove implementation, and a working system does not prove validated performance or impact.
- A target is not an outcome.
- Do not overgeneralise from one example or flatten uncertainty.
- Do not force a rigid methodology or burden the framework with needless process.

12.3 Status ladder for any claim about a capability

Idea → proposed design → prototype → implemented feature → tested capability → deployed system → empirically validated result. If evidence does not establish the status, mark it **unknown**.

12.4 Four categories of knowledge

- **A. Established framework knowledge:** general practice that applies across contexts.
- **B. Source-derived observations:** ideas or examples from supplied material; check whether they are general, project-specific, evidenced, or only assumptions.
- **C. Recommendations:** suggested practices, labelled as such.
- **D. Unresolved questions:** items the material cannot settle; state what would resolve each.

Never promote an item from one category to another without justification.

13. Glossary and End-to-End Reference Map

13.1 Glossary

Term	Meaning
Acceptance criteria	Conditions under which a requirement counts as satisfied
Baseline	The starting level used for comparison
Benchmark	A defined basis for assessing performance
Capability boundary	What an AI system is and is not intended to handle
Decision gate	A point where a defined decision is made on defined evidence

Term	Meaning
Decision record	A short note of a consequential decision and its reasoning
Evaluation dataset	Examples used to assess system behaviour
False positive / false negative	A wrongly triggered detection / a missed real case
Guardrail metric	A measure that must stay within acceptable limits
Human oversight	A defined human role with trigger, owner, information and authority
Hypothesis	A testable prediction, not a finding
MVP	The smallest scope that tests key assumptions or delivers meaningful value
Non-functional requirement	A quality or constraint such as performance, privacy or accessibility
Operational definition	A precise statement of what a metric measures and how
PRD	A focused document on what a product or feature must achieve, with scope and acceptance criteria
Regression testing	Checking that a change has not broken earlier behaviour
Release readiness	Evidence-based confidence that a release is acceptable given its risks
Responsible AI	Practices that tie risks to controls, tests, owners and release criteria
Rollback	Restoring a previous state or disabling a capability after a problem
Target	A desired level of performance, not an achieved result
Traceability	Useful links from intent to requirement, test, evidence and decision

13.2 Reference map

Stage of work	Core question	Key evidence	Main artifact	Feeds back to
Discover	Right problem and users?	Research synthesis, assumptions logged	Discovery and Strategy	Production findings
Define	What must it achieve?	Testable requirements	PRD and backlog	Evaluation, usability
Design	How do people use it, including failures?	Prototype and usability findings	UX and AI behaviour docs	Evaluation
Evaluate	Does it work well enough?	Scenario results and limits	Evaluation and metrics	Design, requirements
Experiment and deliver	Is release justified and controlled?	Experiment results, readiness assessment	Experiment and release plan	Discovery, priorities
Launch	Are controls and rollback ready?	Go/no-go record	Release checklist	Monitoring
Monitor and improve	Is value continuing?	Signals, incidents, feedback	Monitoring and improvement plan	Every earlier stage

14. Source Status, Limitations and Open Questions

This register states honestly what the package was built from and what remains unresolved. It does not claim exhaustive source reconciliation.

Item	Status	What would resolve it
Document 1 ends mid-sentence in Study 05, section 7.7	Truncated	Resend the rest of Document 1

Item	Status	What would resolve it
Study 06 and Study 07 text from Document 1	Not received; sections 8 and 9 are synthesised from the master instruction and Document 2	Send the original Studies 06 and 07 so they can be reconciled
Last item in section 7.7 (measuring activity instead of outcomes)	Completed by inference	Confirm against the original sentence
Document 3 ends mid-sentence in its technical requirements ("If the implementation uses bro...")	Truncated; the remaining technical guidance was not received	Resend the end of Document 3
The Illustrative AI Customer Support Project source material	Not supplied; only the master instruction's topic list was available	Provide it if project-specific observations are wanted; none are asserted here
Metric reconciliation (for example resolution time versus handling time)	No project metrics supplied, so nothing to reconcile	Provide the metric definitions and targets from the source documents
Research provenance of any personas or interviews	Not assessed; none supplied	Provide the research material
Implementation and validation status of any capability	Unknown for the illustrative project; this booklet makes no claims about it	Provide evidence of what was built and tested
Mapping of study areas to layers (section 2.2)	Proposed resolution, not a confirmed one	Review against the original source

Who maintains this booklet. Treat it as a living document: revise it when the missing sections arrive or when new evidence from real projects changes a practice.